
Bài 6 – Phân tích và Quản lý đầu tư nâng cao

Calculate and compare
rates of return

Considering both risk
and return

Considering both risk and return



Why different levels of risk and profitability?



Bonds

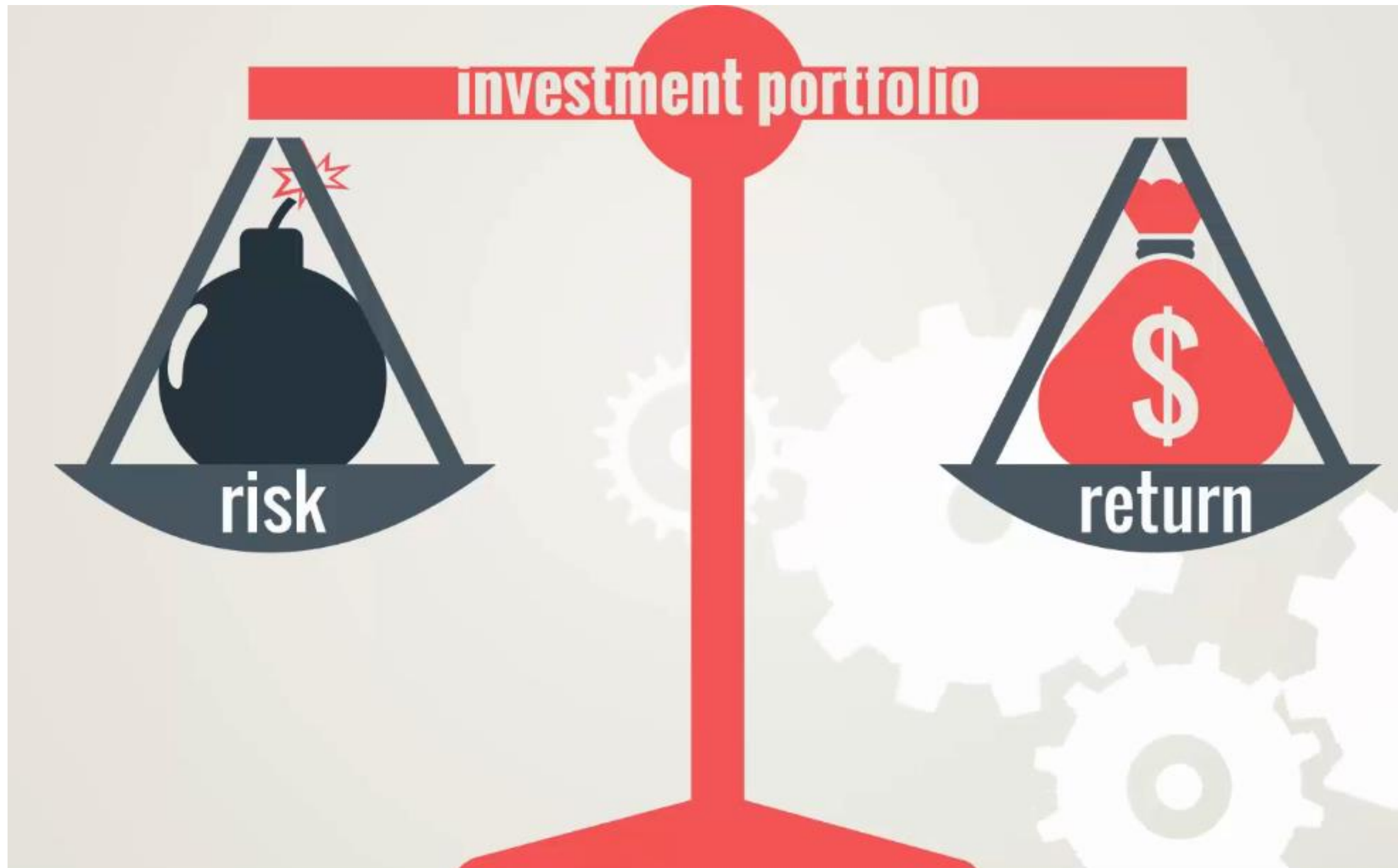
- * 3 % return rate**
- * low risk**
- * initial investment plus interest**



Stock

- * 6 % return**
- * high fluctuations**
- * price change**
- * higher returns are at the price of higher risk**

The art of finance is all about...



Quiz

How would you comment on the statement “There is absolutely no risk involved when investing in Government bonds”?

- Always True
- Sometimes True
- False
- Depends on the level of the expected rate of return.

Calculating a security's rate of return

Calculating a security's rate of return



Bought

1 share

Buying price

\$105

End of year price

\$116

Profit = \$11

Is this a lot?

Calculating a security's rate of return

Is this a lot?

Profit = \$11

Compare



VS



VS



\$116
per share

\$944
per share

\$835
per share

Compare and Calculation



VS



VS



\$116
per share

\$944
per share

\$835
per share

- ? Calculate a percentage rate of return of the investment
- ✓ Compare to other investments

$$\text{rate of return} = \frac{\text{ending price} \text{ minus } \text{beginning price}}{\text{beginning price}}$$

Simple rate of return

$$10.5\% = \frac{\text{\$116} \text{ minus } \text{\$105}}{\text{\$105}}$$



Simple rate of return

$$12.4\% = \frac{\text{\$116}^{+\$2} \text{ minus } \text{\$105}}{\text{\$105}}$$

Logarithmic rate of return

$$10.0\% = \log \frac{\$116}{\$105} = \log \$116 \text{ minus } \log \$105$$

Simple rate of return

When dealing with multiple assets over the same timeframe

Logarithmic rate of return

When you make calculations about a single asset over time

Logarithmic rate of return

$$10.0\% = \log \frac{\$116}{\$105} = \log \$116 \text{ minus } \log \$105$$

Simple rate of return

When dealing with multiple assets over the same timeframe

Logarithmic rate of return

When you make calculations about a single asset over time

Timeframe

Investments with different holding periods should not be compared



$$\text{annual return} = [(\text{daily return} + 1)^{365}] * 100 - 1$$

Converts daily, monthly, and quarterly returns to yearly

Rates of return



Quiz

Which of the following is correct?

- We should always use simple returns and never use logarithmic returns
- It is best to use log returns for calculations when interested in multiple assets over the same timeframe
- Log returns are preferable when you make calculations about a single asset over time
- Simple returns can be applied in all situations. Log returns can be applied only in some specific cases.

Calculating return of multiple investments

Calculating the rate of return of a portfolio

0,318524		25%		
0,153581		25%	$25\% * 0,318524 = 0,079631$	
			$25\% * 0,153581 = 0,038395$	
0,116717		25%	$25\% * 0,116717 = 0,029179$	
			$25\% * 0,126224 = 0,031556$	
0,126224		25%		

$= 0,178761$
rate of return!

add up

rate of return for a security * weight in portfolio

Calculating return of multiple investments

Calculating the rate of return of a portfolio

with differentially weighted securities

0,318524  40%

0,153581  40%

0,116717  15%

0,126224  5%

$$40\% * 0,318524 = 0,127410$$

$$40\% * 0,153581 = 0,061432 = 0,212661$$

$$15\% * 0,116717 = 0,017508$$

$$5\% * 0,126224 = 0,006311$$

rate of return for a security * weight in portfolio

Still the same!

Quiz

What do we need to calculate the rate of return of a portfolio?

- The weights of the securities
- The average return of each security
- The risk value of each security
- Both the weight and the average return of each security (Correct)

Popular stock indices that can help us understand financial markets

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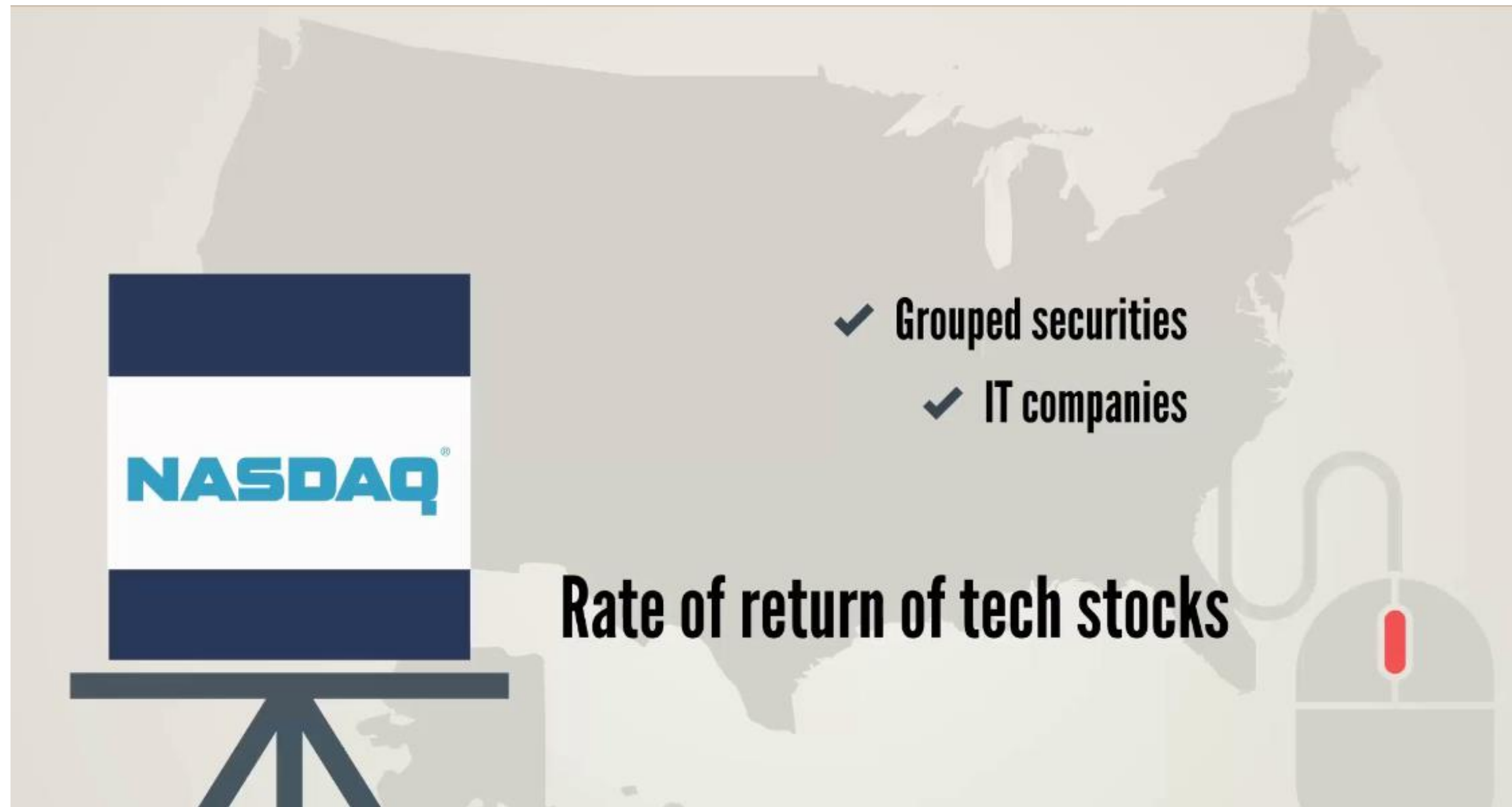
- ✓ 500 largest listed companies
- ✓ Diverse constituency
- ✓ Market-cap-weighted

True approximation of the US stock market

- ✓ 30 large public stocks
- ✓ one of the historically oldest indices

NOT a true representation of the US stock market





And some foreign stock indices...

FTSE

DAX30
UNSER BESTSELLER

NIKKEI

 上海證券交易所
SHANGHAI STOCK EXCHANGE

Germany

Japan

China

Global indices?

Yep.

MSCI



Includes stocks from all developed markets in the world



Why know about
stock indices?

Stock indices are an excellent comparator to
understand how your own stocks are performing

They indicate what to expect if you invest in a
diversified portfolio

**THANKS FOR YOUR
ATTENTION!**